COMPARING THE LOCALIZED TOPOS SH(C/K) WITH THE SLICE TOPOS SH(C)/F(K)^#

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the product decomposition of Sh(C/K) and of the slice topos (source lines 3742-3746) is the exact componentwise mechanism also used on card L46179 (prepare-bbd-glueing); the sheaf identification F(K)^# = coprod h_{U_i}^# is read off from the definition of F; no verification is omitted.

Editorial note. The label lemma-localize-compare names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Semi-representable objects and the functor F. Same chapter, Section \label{section-semi-representable} and Hypercoverings, Definition \label{hypercovering-definition-SR-F}. Statement lines 3736-3742: K in SR(C); F(K)^# is the sheaf associated to the coproduct of the represented objects.
- Localization j_! for a single object. Sites, Lemmas \label{sites-lemma-essential-image-j-shriek} and \label{sites-lemma-compute-j-shriek-restrict}. Proof lines 3747-3752: (1) and (2) follow from the corresponding single-object statements applied componentwise.
- Relocalization morphisms for a morphism of semi-representable objects. Sites, Lemma \label{sites-lemma-relocalize-explicit}. Proof lines 3754-3772: (3) follows by applying this lemma to each re-localization morphism C/U_i -> C/V_{alpha(i)}.

Source reference. Original definition 01G1: sources/stacks-chow/tags.txt, source lines 1890–1890. Prerequisite pointer invoked at proof line(s) 3742; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EE: sources/stacks-chow/tags.txt, source lines 4453–4453. Prerequisite pointer invoked at proof line(s) 3764; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 00Y1: sources/stacks-chow/tags.txt, source lines 1239–1239. Prerequisite pointer invoked at proof line(s) 3763; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04IL: sources/stacks-chow/tags.txt, source lines 5907–5907. Prerequisite pointer invoked at proof line(s) 3768; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site.*

(1) *For K in $SR(\mathcal{C})$ the functor $j_!$ gives an equivalence $Sh(\mathcal{C}/K) \to Sh(\mathcal{C})/F(K)^\#$ where F is as in Hypercoverings, Definition 01G1.*
(2) *The functor $j^{-1} : Sh(\mathcal{C}) \to Sh(\mathcal{C}/K)$ corresponds via the identification of (1) with $\mathcal{F} \mapsto (\mathcal{F} \times F(K)^\# \to F(K)^\#)$.*
(3) *For $f : K \to L$ in $SR(\mathcal{C})$ the functor f^{-1} corresponds via the identifications of (1) to the functor $Sh(\mathcal{C})/F(L)^\# \to Sh(\mathcal{C})/F(K)^\#$, $(\mathcal{G} \to F(L)^\#) \mapsto (\mathcal{G} \times_{F(L)^\#} F(K)^\# \to F(K)^\#)$.*

Proof. Observe that if $K = \{U_i\}_{i \in I}$ then the category $Sh(\mathcal{C}/K)$ decomposes as the product of the categories $Sh(\mathcal{C}/U_i)$. Observe that $F(K)^\# = \coprod_{i \in I} h_{U_i}^\#$ (coproduct in sheaves). Hence $Sh(\mathcal{C})/F(K)^\#$ is the product of the categories $Sh(\mathcal{C})/h_{U_i}^\#$. Thus (1) and (2) follow from the corresponding statements for each i, see Sites, Lemmas 00Y1 and 03EE. Similarly, if $L = \{V_j\}_{j \in J}$ and f is given by $\alpha : I \to J$ and $f_i : U_i \to V_{\alpha(i)}$, then we can apply Sites, Lemma 04IL to each of the re-localization morphisms $\mathcal{C}/U_i \to \mathcal{C}/V_{\alpha(i)}$ to get (3). $\square$